# Lecture 6 (9/25)

*Variance, Normal models and Linear Regression*

# Variance and standard deviation

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**Related quantities:**

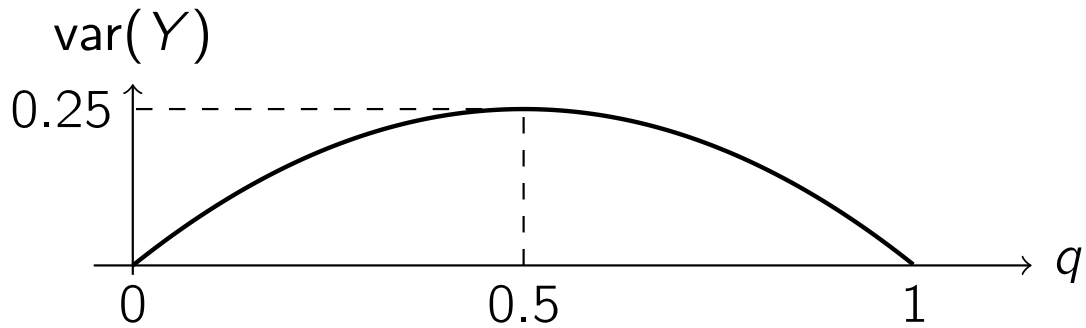
- The standard deviation is $\sigma = \sqrt{\text{var}(Y)}$
- The coefficient of variation is $\text{CV}_Y = \sigma/\mu$ where $\mu = E[Y]$.

## Question

Let $Y \sim \text{Bernoulli}(q)$. Compute $\text{var}(Y)$.

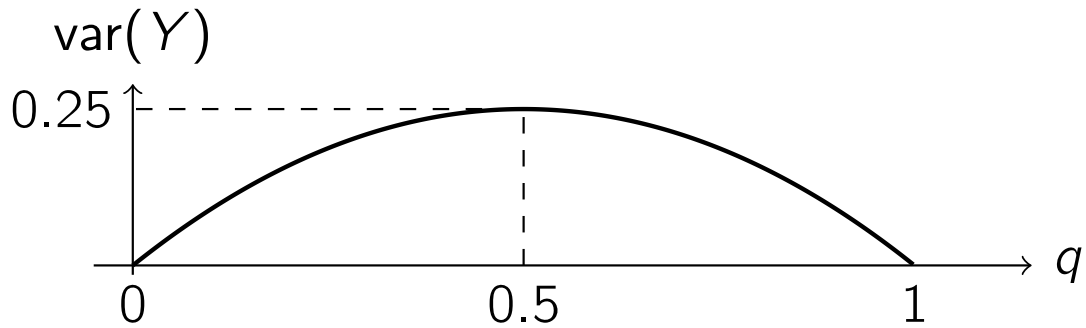
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Let $Y \sim \text{Uniform}(0, L)$. Compute $\text{var}(Y)$ and CV.

# The Normal distribution

$Y \sim \text{Normal}(\mu, \sigma^2)$ has density

$$g(y) = \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-(y-\mu)^2/2\sigma^2},$$

with $E[Y] = \mu$ and $\text{var}(Y) = \sigma^2$.

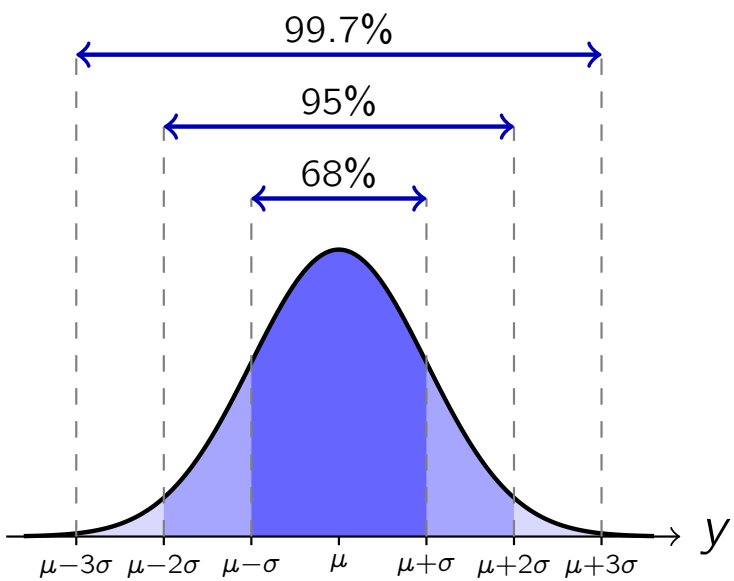
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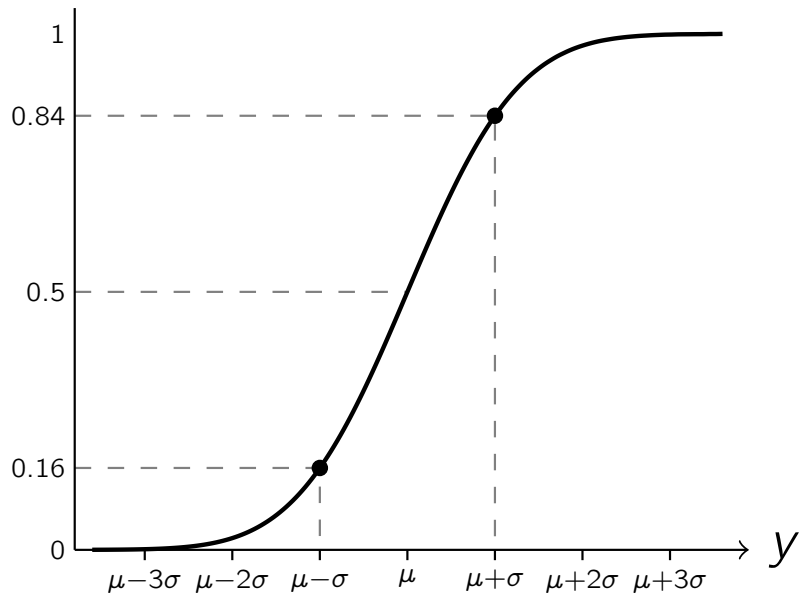
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density $g(y)$




CDF $P(Y \leq y)$



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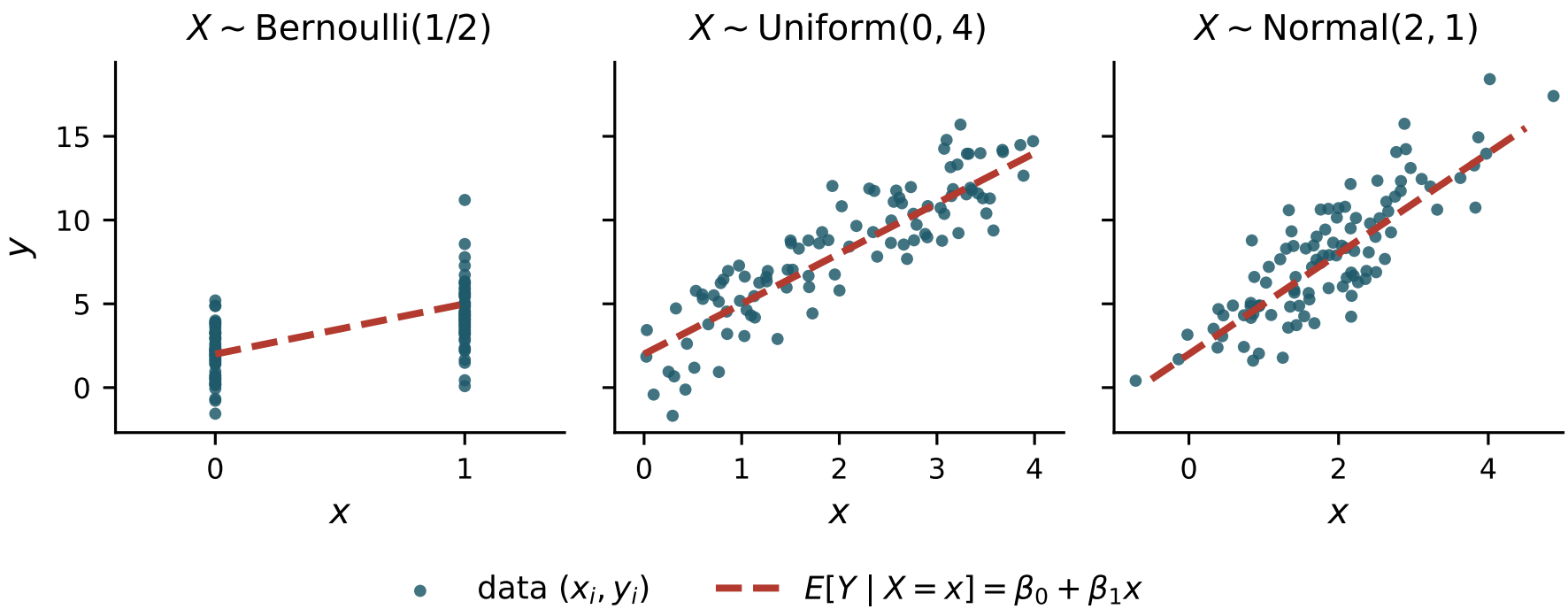
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- $X$ is the predictor
- $Y$ is the response variable
- $\beta_1$ is called the regression coefficient of $X$.
- The variable $\epsilon$ may be referred to as an exogenous noise source. Exogeneity means this noise is independent of $X$.

# What data from this model looks like

Data simulated from $Y \mid X \sim \text{Normal}(\beta_0 + \beta_1 X,\ \sigma_\varepsilon^2)$



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3) Homoskedasticity: $\mathsf{var}(Y \mid X = x)$ does not depend on $x$.

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This formula connects the regression coefficient to the covariance (and hence to expectations we can compute from data).